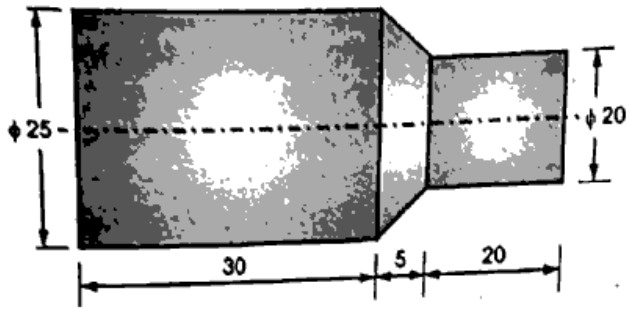


Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Differentiate clearly between NC and CNC systems. Classify NC/CNC systems in detail.	CO1	6
B) Write in short about recirculating ball screws used in CNC machines.	CO1	6
C) What are the different feedback devices used in CNC machines? Explain any two in detail.	CO1	6
Q.2 Solve Any Two of the following.		12
A) Prepare a part program to machine the work piece as shown in the Figure.	CO2	6
		
B) Describe Automatic Tool Changers (ATC) used in CNC machines.	CO2	6
C) Explain the work holding devices used in CNC machines.	CO2	6
Q. 3 Solve Any Two of the following.		12
A) Discuss Electro-Chemical Machining process in detail.	CO3	6
B) Compare Laser Beam Machining with Electron Beam Machining.	CO3	6
C) Explain the process of Abrasive Jet Machining in detail.	CO3	6

Q.4	Solve <i>Any Two</i> of the following.		12
A)	Explain the basic principles involved in thermal spraying with the help of schematic diagrams. What are the different types of thermal spraying methods?	CO4	6
B)	Differentiate clearly between Chemical Vapour Deposition and Physical Vapour Deposition techniques. Explain the basic principles of both the processes using schematic diagrams.	CO4	6
C)	Briefly describe the electroplating and electroless plating processes.	CO4	6
Q. 5	Solve <i>Any Two</i> of the following.		12
A)	Explain the steps involved in rapid prototyping process.	CO5	6
B)	Describe 3-D printing process in detail	CO5	6
C)	Explain the LIGA micro-fabrication method with the help of schematic diagrams.	CO6	6

***** End *****